

1, Explain uses of ping command in network ?

- Used to check whether an another device is reachable in a network and measure response time.
- Uses ICMP message
 - ICMP Echo request
 - ICMP Echo reply

2, Explain netbios protocol ?

- A Protocol / Service that allows Computer in a LAN Communicate and identify each other using Computer names

3, Explain SMB protocol ?

- Server Message Block
- A Client-Server Communication protocol used for sharing access to files, printer and Serial ports on a network.

4, Explain ping process from 50.0.0.10 to 50.0.0.20 ?

- OSI Layer Encapsulation
- Ping uses ICMP (Internet Control message protocol)
- Layer 7 - Application protocol - User Started a ping ICMP create Echo request. Data Size is 32 bytes
- Layer 6 - Presentation Layer - formats or Convert to Standard format
- Layer 5 - Session layer - Establish and manage communication & maintain session
- Layer 4 - Transport layer - Called as Segment
- Layer 3 - network layer - Adds the IP address. Called as Packet.

- Layer 2 - Ethernet layer Data Link layer.
 - add Ethernet header. Contains Src mac - dest mac.
 - Ethernet Protocol doesn't know the dest ip mac so using ARP
 - ARP - used to getting dest mac using dest IP address.
 - done two thing - ARP request ARP reply
- Physical Layer - Layer 1
 - Converts Frame into bits and send through a wired or wireless medium.